

Q4(a)(i)

By Ohm's Law, $V = IR$

$$R = \frac{V}{I} = \frac{5.90}{0.964} = 6.12 \Omega$$

Q4(a)(ii)

Let the third resistor be C. Resistance of C, $R_c = \frac{6.12}{2} = 3.06 \Omega$

Combined resistance of B and C in parallel $R_{BC} = \left(\frac{1}{6.12} + \frac{1}{3.06} \right)^{-1} = 2.04 \Omega$

Total resistance in the 2nd setup is hence $R_{total} = 2.04 + 6.12 = 8.16 \Omega$

Also, current through resistor A, $I = \frac{V}{R} = \frac{8.48}{6.12} = 1.3856 \text{ A}$

Using $\mathcal{E} = V + Ir$, where V is the external p.d. and r is the internal resistance, for the new setup, we get: $\mathcal{E} = IR_{total} + Ir$

$$\mathcal{E} = 1.3856 \times 8.16 + 1.3856r = 11.31 + 1.3856r \text{ --- Equation (1)}$$

Using $\mathcal{E} = V + Ir$ for the previous setup, we get

$$\mathcal{E} = (5.90 \times 2) + 0.964r = 11.80 + 0.964r \text{ --- Equation (2)}$$

Solving the simultaneous equation, by letting Equation (1) = Equation (2)

$$11.31 + 1.3856r = 11.80 + 0.964r$$

$$1.3856r - 0.964r = 11.80 - 11.31$$

$$0.4216r = 0.49$$

$$\mathbf{r = 1.16 \, \Omega}$$

Q4(b)(i)

When the temperature across the thermistor increases, the resistance of the thermistor decreases.

This causes the resultant resistance of the parallel arrangement of B and the thermistor to decrease.

By the potential divider principle, p.d. across A increases (use $V_A = \frac{R_A}{R_A + R_{combined}} \times emf$).

[must show the equation to get the mark. Alternatively, can explain current increases as total resistance falls, and thus by $V = IR$, the p.d. across A increases.]

Q4(b)(ii)

Connect across resistor A in parallel.

Q5(a)